Yifei Zhang

Ph.D. Student, Department of Biostatistics, Yale School of Public Health
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Education

Yale University 2025 – present
Ph.D. in Biostatistics, Yale School of Public Health. Advisor: Prof. Yize Zhao

University of North Carolina at Chapel Hill 2023 – 2025
M.S. in Statistics and Operations Research. Advisor: Prof. Zhengwu Zhang

University of Liverpool 2019 – 2023
B.S. in Applied Mathematics, First Class Honours

Research Interests

Deep generative models, normative modeling, disease progression modeling, multimodal neuroimaging, and brain connectomics.

Publications

1. Gao, H., Ham, D., **Zhang, Y.**, Tian, X., Ding, S., Zhao, Y.* and Li, T.* (2026). Orthogonal sparse subgraph alignment for structure–function coupling in brain networks. *Conference on Neural Information Processing Systems (NeurIPS)*, accepted.

2. Ding, S., **Zhang, Y.**, Tian, X., Guo, Y. and Zhao, Y. (2026). ELPAC: Endpoint-anchored latent progression with stage-varying multimodal coordination. *Conference on Neural Information Processing Systems (NeurIPS)*, accepted.

3. Tian, X., Guo, Y., Ding, S., **Zhang, Y.**, Esserman, D. and Zhao, Y. (2026). A Bayesian deep learning framework for brain dynamic functional connectivity linked with cognition. *Biometrics,* 82(4), ujag174. doi:10.1093/biomtc/ujag174

Preprints

4. **Zhang, Y.**, Liu, M. and Zhang, Z. (2026). Attention-based variational framework for joint and individual components learning with applications in brain network analysis. *arXiv preprint* arXiv:2601.17073. Under review.

Research Experience

Ph.D. Research, Yale University 2025 – present
Advisor: Prof. Yize Zhao. Deep generative models and statistical learning for multimodal neuroimaging, brain connectomics, and disease progression.

M.S. Research, University of North Carolina at Chapel Hill 2023 – 2025

- *Multimodal disentangled network VAE* (with Prof. Meimei Liu and Prof. Zhengwu Zhang), 2024 – 2025. Graph variational autoencoders that separate shared and modality-specific components of structural and functional connectivity and predict traits from connectomes.

- *Surface-based connectivity integration (SBCI) for the Human Connectome Project* (with Prof. Zhengwu Zhang), 2024 – 2025. Preprocessed HCP-Aging and HCP-Development diffusion MRI, T1, and resting-state fMRI (FSL, MRtrix3) and extended the SBCI pipeline to generate connectivity under multiple atlases.
- *EPIC: Prediction of Child Behavior Checklist scores* (with Prof. Guorong Wu and Prof. Zhengwu Zhang), Summer 2024. Combined connectome features (node2vec, JIVE) with XGBoost and Shapley-value feature selection to predict CBCL labels in the ABCD study.
- *Structural connectomes for trait prediction in adolescents* (with Prof. Zhengwu Zhang), Spring 2024. Compared PCA-based SVM, logistic regression, and stacking models for predicting traits such as BMI from structural connectomes.

Talks

"Connecting questions to questions: How to translate real-world questions into deductible insights." TCM Conference, Durham, NC — 2024

Teaching

Teaching Assistant, BIS 623 Advanced Regression Models, Yale University — Fall 2026

Teaching Assistant, DATA 120 Ethics of AI and Data Science, UNC Chapel Hill — 2024 – 2025

Teaching Assistant, STOR 113 Decision Models for Business and Economics, UNC Chapel Hill — Spring 2024

Teaching Assistant, STOR 445 Stochastic Modeling, UNC Chapel Hill — Fall 2023

Honors and Awards

SDSS Departmental and In-State Tuition Award (fully funded M.S. student), UNC Chapel Hill — 2024

First Class Honours in Applied Mathematics, University of Liverpool — 2023

Skills

Programming	R, Python (PyTorch), MATLAB, Java, LaTeX
Neuroimaging	FSL, MRtrix3, FreeSurfer
Tools	Git, Linux
Languages	Chinese (native), English

Last updated: October 2026